



To Renounce Liberty? A Public-Goods Experiment on (Un)Democratic Reversal

Research Plan for Ethical Review

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Abstract

Democratic backsliding often proceeds through lawful democratic choices that weaken citizens' control over what happens next. This project asks whether citizens vote to restore an approval right they supported surrendering once its crisis-based justification recedes. In a one-session public-goods experiment with 300 participants in Finland, five-person groups decide whether an unknown leader's proposal requires citizen approval or takes effect directly. Acting without approval initially combines a higher fund rate with assured implementation of the leader's equal allocation. Structural recovery improves public-service conditions and raises the approval-constrained rate to match direct implementation, while the leader's powers and decision rules remain unchanged. Comparing randomly assigned recovery and persistence pools identifies whether recovery causes previous supporters of direct implementation to vote for group approval. The experiment captures a demand-side microfoundation of executive aggrandizement: citizens can use one democratic decision to weaken a later constraint and another to vote to take that control back.

Experiment prototype (production phase):
<https://panel-lab-experiment.onrender.com>

1 Research Question and Contribution

Democratic backsliding depends not only on leaders who weaken constraints but also on citizens who authorize them. This project studies a narrow demand-side microfoundation of executive aggrandizement: the citizen-level authorization through which a lawful democratic decision weakens a constraint on one-person authority. It asks: *after a crisis recedes, do citizens who supported decisions without their approval vote to restore that right?* This within-person change in institutional support is the **(un)democratic reversal** the project seeks to explain.

We build on the democratic backsliding and public-good games literatures. Contemporary backsliding often advances without ending elections: citizens may support elected officials or lawful rule changes that reduce the approval, oversight, or veto constraints governing later decisions (Bermeo 2016). The politically relevant step is therefore not necessarily an explicit vote against democracy, but a *democratic* vote that makes subsequent authority *less* constrained. Threat and policy benefits can make citizens more tolerant of such constraints being weakened (Hetherington and Suhay 2011; Goodman 2021; Saikkonen and Christensen 2023), while laboratory evidence shows that gridlock can increase support for executive special powers (Forteza, Mussio, and Pereyra 2024).

This literature mainly explains movement toward less constrained authority. Our contribution is to follow the reverse movement: once the crisis-based benefit recedes, do the same citizens vote to restore the approval right they previously supported surrendering? Observing that second choice turns reversal into a consequential decision over an institutional constraint rather than a stated evaluation of democracy.

Public-good games supply the economic reason for weakening approval. Voluntary provision is vulnerable to free riding—keeping one’s points while others fund the shared service—and declining cooperation, whereas centralized enforcement can coordinate contributions (Fehr and Gächter 2000; Fischbacher and Gächter 2010; Hamman, Weber, and Woon 2011). Experiments also show that voting over institutions changes their performance (Dal Bó, Foster, and Putterman 2010; Sutter, Haighner, and Kocher 2010; Markussen, Putterman, and Tyran 2014). Our experimental design joins these literatures by making citizen approval both institutionally meaningful and economically costly.

In each round, five citizens begin with 20 points each and vote between two rules for managing a shared public-services fund. Under both rules, the software selects one of the five citizens uniformly at random only after the vote, so each has the same one-in-five chance of leading that round. The leader proposes how many points every citizen places in the fund and whether any collected points move to the leader’s payoff. Fund points are multiplied and the resulting group return is divided equally.

Under the approval-constrained rule (D), at least three of the other four citizens must approve the complete proposal. Rejection returns the group to voluntary individual allocations. Under automatic implementation (A), the same proposal takes effect without that approval vote. A matching pool contains 10 or 15 citizens and is reshuffled into two or three temporary groups of five before each round. Thus, the democratic ballot selects an approval rule, not a known leader or proposal.

Everyone begins with a public-service crisis: *“Parliament has failed to pass a budget for eight months, healthcare waiting times have doubled, and planning for public services has stalled.”* During Block 1, each fund point creates 1.50 group points under D and 2.50 under A. Direct implementation therefore has a one-point productivity premium and guarantees that the leader’s proposal is implemented. Afterward, each matching pool receives **institutional recovery** or **institutional persistence**. Recovery is one structural package: public-service conditions improve and the D rate rises to 2.50. Persistence retains the crisis and the 1.50 rate. The A rate and decision rule remain fixed.

The payoffs make the choice consequential; **the allocation of decision rights connects it**

to democratic backsliding. A citizen who votes for A supports surrendering an approval vote before knowing the leader or proposal. A later vote for D supports restoring that safeguard after the crisis-based productivity advantage disappears. *The experiment therefore observes whether citizens vote to take back a specific constraint on authority, rather than inferring reversal from a survey declaration.*

In simple terms

Five citizens vote before seeing who will lead or what the leader will propose. Under D, three of the other four citizens must approve the proposal; rejection returns everyone to individual fund allocations. Under A, the proposal takes effect directly. In Block 1, these methods use rates of 1.50 and 2.50, respectively.

After Block 1, recovery improves public-service conditions and raises D to 2.50; persistence leaves the crisis and rate gap in place. Among previous A supporters, a later D vote supports restoring the approval right. Comparing recovery with persistence identifies how much the structural recovery package changes that reversal.

Central insight

A decision can be democratically authorized while weakening the democratic constraints that govern what happens next. This resembles contemporary backsliding, which often proceeds through elections and formally lawful institutional changes rather than an outright abolition of democracy (Bermeo 2016).

In the experiment, the first majority vote is democratic. A citizen who chooses A votes to let an unknown leader's later allocation and transfer take effect without the other citizens' approval. That vote removes a specific constraint while leaving the institutional ballot itself intact. A later vote for D supports restoring the approval right the citizen previously voted to surrender.

What matters is that citizens use one democratic decision to weaken their control over subsequent decisions—and can later vote to take that control back.

2 Theory

2.1 Crisis and the productivity premium

A crisis can make action without approval attractive when ordinary procedures produce delay or failed coordination. The experiment gives this argument material consequences. In Block 1, proposals requiring approval (D) use a 1.50 fund rate, while proposals implemented directly (A) use 2.50. Under D, rejection also returns the group to voluntary contributions and possible free riding. Direct implementation therefore offers both a productivity premium and assured implementation of the leader's equal allocation (Fehr and Gächter 2000; Fischbacher and Gächter 2010; Hamman, Weber, and Woon 2011).

The personal-transfer option makes the institutional constraint consequential. The leader may propose moving up to 20 fund points to their own payoff under either procedure. Under D, the other citizens see that transfer before implementation and can reject it; under A, they observe it only after it has taken effect. The experiment therefore concerns approval over discretion, not the prevalence of corruption in Finland.

H1: Authorization under crisis. During Block 1, citizens will increasingly support direct implementation (A) when its productivity and coordination advantages produce better outcomes; larger observed transfers will reduce subsequent support.

2.2 Citizen authorization and democratic constraint

The institutional ballot occurs before leader selection and before the proposal. A vote for A therefore authorizes a rule, not a known person or action: the citizen relinquishes participation in approving the next binding decision. A vote for D retains that approval right. **Thus, our design provides an abstract microfoundation of *executive aggrandizement*, in which a *democratic choice weakens the constraint governing what follows*** (Bermeo 2016; Forteza, Mussio, and Pereyra 2024).

H2: Approval and discretion. Approval will decline as proposed personal transfers increase. Anticipating rejection, leaders operating under D will propose smaller transfers than leaders whose proposals take effect directly. Because groups choose the institutional rule themselves, these patterns describe behavior within the chosen rules rather than a separate randomized effect.

2.3 Recovery and democratic reversal

Recovery is assigned at the matching-pool level. Pool g contains 10 or 15 participants, who are reshuffled into two or three temporary groups of five before each round; within each group, three votes produce a majority. The entire pool receives the same Block-2 condition: $T_g = 1$ denotes recovery and $T_g = 0$ persistence. Let m_{gb} denote the approval-constrained fund multiplier in pool g and block b . The productivity premium for direct implementation is

$$\delta_{gb} = 2.50 - m_{gb} \tag{1}$$

Everyone faces $\delta_{g1} = 2.50 - 1.50 = 1$ in Block 1. Recovery is deliberately implemented as one structural package: participants learn that public-service conditions improved and m_{g2} rises to 2.50, setting $\delta_{g2} = 0$. Persistence reports that the crisis continues and keeps $m_{g2} = 1.50$, so $\delta_{g2} = 1$. The leader-selection rule, proposal set, approval threshold, rejection-to-voluntary-allocation rule, and A rate remain fixed.

The recovery–persistence comparison identifies the total causal effect of structural recovery on (un)democratic reversal. Among final-round Block-1 supporters of A, the analysis compares later D votes in recovery pools ($T_g = 1, \delta_{g2} = 0$) with those in persistence pools ($T_g = 0, \delta_{g2} = 1$). The public-service description and production rate jointly define recovery: the text states what changed, while the rate makes that change materially consequential inside the game. The estimand therefore concerns their combined effect rather than separate framing and payoff effects. Persistence supplies the counterfactual in which both the crisis and premium remain.

H3: Immediate restoration. Among participants who vote for A in Block 1’s final round, recovery will increase the probability of voting for D in Block 2’s first round.

H4: Sustained reversal. Among the same participants, recovery will increase the probability of voting for D in Block 2’s final round.

3 Experimental Design

3.1 Overview

Each laboratory cohort completes one session lasting approximately 60 minutes, with two ten-round blocks. Subject to ethical review in September 2026, data collection is planned for 1–31 October 2026 at the UTU Choice Lab decision-making laboratory and may also be conducted at Tampere University’s Decision-making Laboratory (DMLab) to accelerate recruitment. UTU

Choice Lab has 24 regular seats plus one accessible seat, while DMLab has 20; production rosters are capped at 20. Participants remain in matching pools of 10 or 15 and are reshuffled into anonymous five-person groups before every paid round. After Block 1, each entire pool is assigned to recovery or persistence, learns whether public-service conditions improved or remained strained, and sees its Block-2 fund rates. Participants therefore interact repeatedly without forming stable five-person teams. In the figures, p_i identifies participant i .

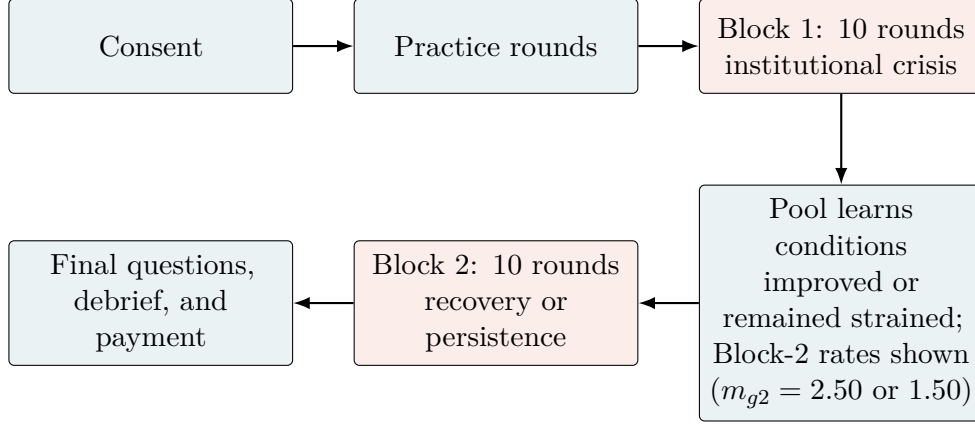


Figure 1: One-session sequence. Each paid round includes an institutional ballot, leader proposal, implementation, feedback, and rematching.

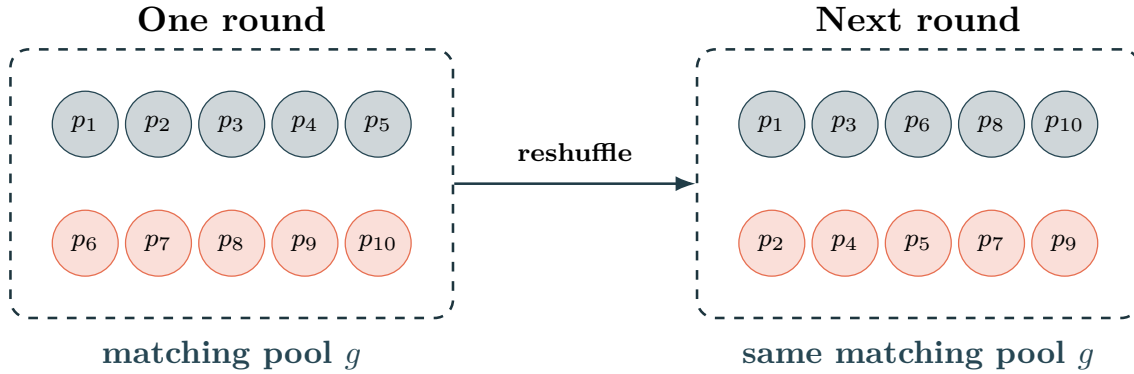


Figure 2: Anonymous rematching within a matching pool (ten-person example).

Notes: Each participant p_i remains in pool g . A ten-person pool forms two temporary groups per round; a fifteen-person pool forms three. Recovery or persistence applies to the whole pool in Block 2.

In simple terms

Matching pools permit interaction without fixing the same five-person team. Round-by-round reshuffling limits stable reputations, collusion, and retaliation, while keeping participants inside one treatment environment. Ten-person pools are used whenever the completed cohort can be divided that way; one fifteen-person pool accommodates a remaining set of five while retaining rematching.

Reversal is measured for individuals, but treatment (T_g) is assigned to pools. Statistical uncertainty is therefore clustered by pool; temporary groups organize play but are not treatment units. Ten-person pools remain the default because they yield more independent treatment assignments for a fixed sample.

Table 1: Session formation from the completed laboratory roster

People who arrive	Matching pools	Five-person groups per round	Participate	Released
20	10 + 10	$2 + 2 = 4$	20	0
16–19	15	3	15	1–4
15	15	3	15	0
11–14	10	2	10	1–4
10	10	2	10	0
Fewer than 10	—	—	0	All

Notes: A matching pool is not a playing group. Each ten-person pool forms two temporary five-person groups per round; a fifteen-person pool forms three. At 20 participants, two separate ten-person pools each form two groups. Groups dissolve and are reshuffled after every round; participants never enter or change pools after paid play begins.

3.2 The institutional-choice public-good game

Each citizen starts with 20 points and privately votes between an approval-constrained rule (D) and automatic implementation (A). Let $V_{i,b,r} \in \{D, A\}$ denote citizen i 's ballot in block b and round r . Three votes select the binding rule. The software then selects each group's leader by an independent uniform draw among its five citizens. Selection occurs under both rules and only after the institutional ballot, giving every citizen the same ex ante chance in every round without making past service predict the next draw.

Table 2: Implemented design parameters

Feature	Implemented value
Laboratory session	One synchronized visit, approximately 60 minutes
Paid rounds	10 per block (20 total), preceded by unpaid practice
Group and matching pool	5-person groups; stable 10- or 15-person matching pools
Rematching	Anonymous rematching within pool before every paid round
Round endowment	20 points per participant
Approval-constrained multiplier	1.50 in Block 1; 2.50 recovery / 1.50 persistence in Block 2
Automatic-implementation multiplier	2.50 in both blocks and conditions
Approval	At least 3 of the 4 nonleaders; a 2–2 split rejects the proposal
Rejection	Voluntary individual contributions at the approval-constrained rate
Personal transfer	0–20 fund points, limited by the points collected
Performance bonus	Points earned in one randomly selected round from each block
Maximum performance bonus	EUR 5.00
Show-up payment	EUR 10.00
Maximum total payment	EUR 15.00

Selected-round points are converted linearly into money. Because a selected round can yield at most 60 points, the two selected rounds produce a bonus of at most EUR 5. Selecting one round from each block keeps every decision consequential without paying all rounds or encouraging participants to offset one decision with another.

Block 1 — Round 1 of 10

You have 20 points. Points in the **public-services fund** create group points that are shared equally among all five members. Points you do not put in the fund remain yours.

Should the selected person's proposal require approval? Click anywhere on one row.

Method	What happens to the proposal?	Fund rate	If citizens reject
G Group approval required	The other four citizens see the full proposal; at least three must approve	Each point left in the fund creates 1.50 group points	Everyone chooses their own fund allocation at the same rate
D Decision takes effect directly	The proposal is implemented without an approval vote by the other citizens	Each point left in the fund creates 2.50 group points	No rejection stage

Figure 3: Participant screen for choosing whether the leader's proposal requires approval.

Leader proposal. Under either rule, the leader proposes an equal allocation $t \in \{0, \dots, 20\}$ and a personal transfer $\rho \in \{0, \dots, 20\}$, with $\rho \leq 5t$. Let π denote a citizen's payoff in that round. Under A, the proposal takes effect directly, and a nonleader earns

$$\pi_{igbr}^A = 20 - t + \frac{2.50}{5}(5t - \rho), \quad (2)$$

and the leader earns $\pi_{igbr}^{A,L} = \pi_{igbr}^A + \rho$.

Approval-constrained procedure (D). The four nonleaders observe (t, ρ) and vote privately; let $q_{igbr} = 1$ denote approval. The proposal passes when at least three approve. If it passes, the payoffs above apply with m_{gb} replacing 2.50. If it fails, each citizen chooses $c_{igbr} \in \{0, \dots, 20\}$ and earns

$$\pi_{igbr}^D = 20 - c_{igbr} + \frac{m_{gb}}{5} \sum_{j=1}^5 c_{jgbr}. \quad (3)$$

The rejected transfer is not implemented. Everyone then sees the proposal, approval total, implemented allocation, transfer, and own payoff.

In simple terms

Rejection restores the standard **public-goods dilemma**. If one citizen contributes 1 point and four contribute 0, the contributor earns $19 + \frac{1.50}{5} = 19.30$ during the crisis, while each free-rider earns 20.30. Under recovery, these payoffs become 19.50 and 20.50. Contributing helps the group but costs the contributor; direct implementation avoids that risk, while approval lets citizens block an unwanted proposal.

(a) Leader proposal

You are the selected person for this round

Propose how many points every citizen should put in the public-services fund and whether up to 20 fund points should move to your payoff. Your proposal takes effect directly, and the remaining fund uses the 2.50 rate.

How many points should every citizen put in the public-services fund?

How many fund points should move to your own payoff?

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(b) Citizen approval

Approve or reject the proposal

The selected person proposes that every citizen put 10 points in the fund and that 5 fund points move to their own payoff. At least three of the four other citizens must approve. If the proposal is rejected, every citizen will choose their own fund allocation.

Do you approve or reject the selected person's proposal?

☐ Approve the proposal
☐ Reject the proposal

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(c) Proposal, approval, and payoff feedback

Block 1 — Round 2 result

Group choice: Group approval required

Method ballots: Group approval required: 3; Decision takes effect directly: 2.

Selected person's proposed allocation for each citizen	10
Proposed transfer to the selected person's payoff	5
Approval votes among the four other citizens	3 of 4
Approval result	Approved
Total points placed in the public-services fund	50
Fund points moved to the selected person's payoff	5
Points left in the public-services fund	45
Group points received by every member	13.50
Your payoff if this round is selected	23.50

Only group totals are shown. Participants are reshuffled anonymously before the next round.

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Figure 4: Proposal, approval, and round feedback. Entered amounts are illustrative.

How incentives map onto reversal. During Block 1, A combines a 2.50 rate with guaranteed implementation; D uses 1.50 and may return to voluntary contributions after rejection. Recovery jointly reports improved public-service conditions and raises D to 2.50, while persistence retains the crisis and rate gap. The proposal set, approval rule, rejection procedure, and A rule remain unchanged. The recovery–persistence contrast therefore estimates the total effect of this structural package.

From institutional choice to (un)democratic reversal. A citizen who moves from A to D first votes to let an unknown leader’s proposal take effect without approval and later votes to restore that approval. The public-good incentives make this act consequential: A still guarantees that the leader’s equal allocation takes effect after recovery, while D may produce rejection and free riding. A higher switching rate under recovery shows that structural change can cause citizens to withdraw support for a weakly constrained decision rule. Importantly: *reversal is observed as a vote to restore a decision right, not as a declaration of democratic sentiment.*

A simple payoff comparison

Suppose the leader proposes $t = 10$ and $\rho = 0$, and the proposal is approved under D. Holding that decision fixed gives:

	Approval required (D)	Direct implementation (A)
Crisis or persistence	25 points each	35 points each
Recovery	35 points each	35 points each

Recovery makes an approved D proposal equally productive; it does not make it pay more. A still guarantees implementation. **The question is: *when approval becomes equally productive, do citizens vote to restore it?***

Participants report expected payoffs and transfers before the final Block-1 ballot and the first and final Block-2 ballots. Expected payoffs assess whether they understood the rules' relative performance; expected transfers trace beliefs about how leaders will use their discretion.

In simple terms

Five citizens cast private institutional ballots ($V_{i,b,r}$), after which the software selects a leader. That leader proposes (t, ρ) . Under D, three of four nonleaders must approve; rejection activates individual contributions (c_{igbr}). Under A, the proposal takes effect directly. Everyone sees the result and payoff before rematching within pool g .

3.3 Block 1: crisis and leader authorization

After consent, participants practice proposing, approving, rejecting, and making individual allocations after rejection, then answer five comprehension questions. The interface never calls either method democratic or undemocratic.

Block 1 begins with the public-service crisis described above. Each fund point produces 1.50 under D and 2.50 under A. The leader may propose the same allocation and transfer under either rule; what differs is whether the other citizens can reject it.

Each paid round repeats the same sequence: institutional ballot, leader proposal, approval or direct implementation, any individual allocation following rejection, feedback, and rematching. Expectations are elicited before the final ballot, and one Block-1 round is selected for payment.

3.4 Block 2: structural change and reversal

After Block 1, pools are assigned to *recovery* or *persistence*. Recovery jointly reports that the budget passed, waiting times are falling, and planning resumed, and raises the D multiplier from 1.50 to 2.50. Persistence retains the crisis and 1.50 rate. The A rate, leader selection, proposal set, approval threshold, and rejection procedure remain unchanged. Together, the updated description and rate form the structural-recovery treatment.

The first Block-2 ballot measures immediate support for restoring approval; the final ballot measures sustained reversal and is primary. Expectations are elicited before both ballots, and questions about democratic constraints appear only after paid decisions. One Block-2 round is selected for payment.

4 Outcomes, Process Measures, and Estimands

Let $\mathbb{K}\{\cdot\}$ equal 1 when the condition inside braces is true and 0 otherwise. The final Block-1 ballot defines endorsement of automatic implementation as

$$A_{i1} = \mathbb{K}\{V_{i,1,10} = A\}. \quad (4)$$

For the theoretically relevant subgroup with $A_{i1} = 1$, reversal is

$$R_i = \mathbb{K}\{V_{i,2,10} = D\}. \quad (5)$$

The primary estimand is the average treatment effect of structural recovery among Block-1 supporters of automatic implementation:

$$\tau_R = \mathbb{E}[R_i(1) - R_i(0) \mid A_{i1} = 1], \quad (6)$$

where $R_i(1)$ and $R_i(0)$ are participant i 's potential outcomes under recovery and persistence. Recovery includes both improved public-service conditions and the 2.50 D rate; τ_R is the total effect of that package. Because A_{i1} is measured before assignment, subgroup membership is fixed before treatment. The primary test repeatedly reassigns recovery according to the prespecified pool-level procedure to evaluate how unusual the observed contrast is. Because τ_R describes the average eligible citizen, the primary estimate weights eligible participants equally; a secondary analysis gives each pool equal weight. Regression estimates cluster uncertainty by pool and include a prespecified indicator for fifteen-person pools.

Immediate reversal is

$$R_i^I = \mathbb{K}\{V_{i,2,1} = D\} \quad \text{for } A_{i1} = 1. \quad (7)$$

The same comparison applied to R_i^I estimates immediate support for restoration. The final-round outcome R_i remains primary because it records whether that choice persists.

In simple terms

The primary estimand (τ_R) asks: among citizens who voted for direct implementation in Block 1's final round ($V_{i,1,10} = A$; $A_{i1} = 1$), how much more often does recovery ($T_g = 1$), rather than persistence ($T_g = 0$), lead them to vote for group approval in Block 2's final round ($V_{i,2,10} = D$; $R_i = 1$)? Only one potential outcome is observed per citizen, so random assignment identifies the difference. Assignment and inference operate at pool level.

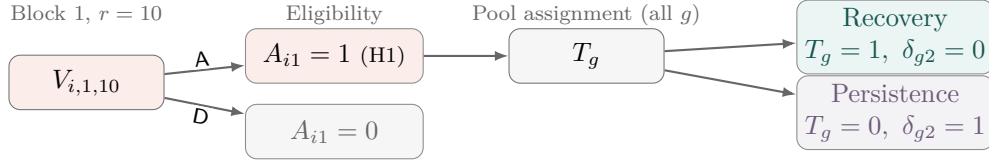
For each leader i , t_{igbr} is the proposed allocation and ρ_{igbr} the proposed transfer. A transfer proposal is $M_{igbr} = \mathbb{K}\{\rho_{igbr} > 0\}$, with intensity $\rho_{igbr}/(5t_{igbr})$ when $t_{igbr} > 0$. Thus, $t = 10$ and $\rho = 5$ proposes moving 10% of the 50 collected points. Approval ballots q_{igbr} , proposal passage, contributions after rejection, and subsequent institutional ballots provide descriptive process evidence. Because groups choose their rule, these measures do not identify separate causal effects of recovery.

Immediate reversal R_i^I records the first response to the structural package, before Block-2 interaction (see Figure 5). The primary outcome R_i records Round 10 after nine completed rounds. Secondary outcomes include late-round automatic-implementation support and realized group-level restoration.

Expected payoffs assess whether participants understood how recovery changed the rules' relative performance, while expected transfers trace beliefs about leader discretion. After all paid choices, eleven process items check perceived crisis severity and recovery, expected institutional performance, and views on approval, equal allocation, and transfers. A separate optional twelve-item background module records age group, gender, education, household financial strain, party voted for in Finland's 2023 parliamentary election, left-right placement, whether democracy is

preferable to other forms of government, satisfaction with democracy in Finland, evaluation of a strong leader unconstrained by parliament and elections, external political efficacy, generalized trust, and willingness to take risks. These prespecified measures capture partisan and ideological trade-offs, distinguish principled democratic support from evaluations of democratic performance, and benchmark cooperative and risk-related predispositions (Graham and Svobik 2020; Saikkonen and Christensen 2023; Claassen 2020; Dohmen et al. 2011; Glaeser et al. 2000). They are collected only after paid decisions and support precision and descriptive heterogeneity analyses; they do not alter the randomized estimand or independently identify a treatment mechanism.

(a) Eligibility and pool assignment



(b) Block-2 paths for $A_{i1} = 1$ under either T_g

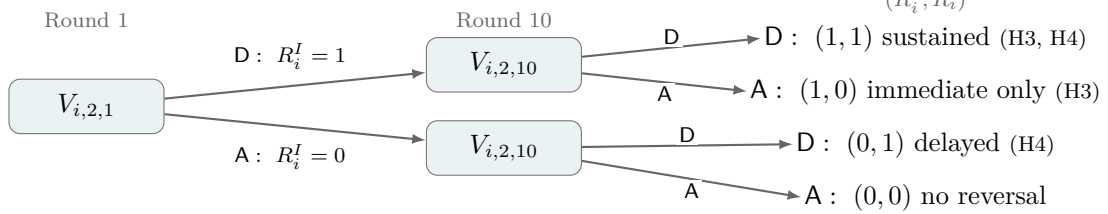


Figure 5: Decision tree for assignment and reversal. Each pool is assigned one value of T_g ; the primary analysis follows participants with $A_{i1} = 1$.

Notes: The displayed ballots are $V_{i,1,10}$ (final Block 1), $V_{i,2,1}$ (first Block 2), and $V_{i,2,10}$ (final Block 2). The ordered pair (R_i^I, R_i) records the two Block-2 outcomes, with 1 = D and 0 = A. Thus, (1, 1) is sustained reversal; (1, 0) immediate only; (0, 1) delayed; and (0, 0) no reversal. Parentheses identify the corresponding hypotheses; H2 concerns transfers and is not shown.

5 Randomization, Sample, and Power

The planned sample comprises 300 completed participants in 28–30 matching pools. Ten-person pools are the default; no more than four fifteen-person pools are used, preserving at least 28 independent treatment assignments. Treatment is assigned only after everyone finishes Block 1. Within each session, pools are ordered by their final-round Block-1 A vote share and paired. One pool in each pair receives recovery and the other persistence, at random; an unmatched pool is assigned by a fair draw. Pairing improves balance in the number eligible to reverse, which is measured before assignment. The maximum payment of EUR 15 implies a participant-compensation ceiling of EUR 4,500 for 300 completed participants.

Power depends on Block-1 endpoint support for A, the persistence-group reversal rate, the similarity of outcomes within pools, and the realised mix of pool sizes. With 300 participants, the design yields 28–30 independent treatment assignments—approximately 14–15 pools per condition—because a larger pool supplies more individual outcomes but still receives only one treatment draw. Repeated rounds improve measurement and reveal trajectories; they do not create additional independent assignments.

Pilot estimates will inform simulations spanning 28–30 pools and the observed number eligible to reverse. The primary randomization test reproduces the actual paired and unmatched assignments. Regression estimates cluster uncertainty by pool, adjust for prespecified pool-size and laboratory-site indicators, and use small-sample inference; an equal-pool-weight analysis checks whether results depend on larger pools contributing more participants. Each oTree session remains a separate interaction and randomization cohort.

Strategic decisions do not time out automatically. The software records response time and flags decisions exceeding 90 seconds, but never generates a ballot or allocation. The laboratory manager monitors the session and prompts delayed participants to complete the current decision so the group can advance.

6 Implementation Plan

The study will be conducted after ethical approval at the UTU Choice Lab decision-making laboratory and may also be run at Tampere University’s Decision-making Laboratory (DMLab) to accelerate recruitment. Adults register through the host laboratory’s participant system and receive its general rules and privacy information. UTU Choice Lab uses UTU RPP/TOKA; DMLab uses ORSEE. At either site, oTree provides project-specific information and records a short affirmative study consent before practice. The laboratory manager then fixes a consenting roster of 10, 15, or 20, forms the matching pools shown in Table 1, and handles any surplus arrivals under the host laboratory’s procedure. No participant enters or changes pools after paid play begins.

The protocol will be piloted before production. A technical pilot will test 10- and 15-person pools, mixed-pool sessions, both Block-2 conditions, wait pages, rematching, response-time flags, browser recovery, exports, and payment calculations. A behavioral pilot will assess comprehension, session length, choice distributions, fatigue, and expected earnings. The protocol, analysis plan, and data dictionary will then be frozen before confirmatory data collection.

Production sessions will last approximately 60 minutes. The laboratory manager will monitor synchronized progress and prompt delayed participants; the software will record delays but will never generate a ballot, proposal, approval decision, or allocation. A structured session log will record late arrivals, early departures, technical interruptions, staff interventions, and payment exceptions. The research team will review the log before applying preregistered data-quality rules.

7 Recruitment, Participant Care, and Resources

The host laboratory will invite adults through its participant system using a neutral description of a paid study of group decision making and public-service provision. Recruitment will not disclose the directional hypotheses. Participation is voluntary and will not affect employment, studies, or access to University services. The fixed participation payment is EUR 10; points earned in one randomly selected round from each block determine a performance bonus of up to EUR 5.

Foreseeable risks are minimal. Participants may experience mild discomfort with unequal allocations, the fictional crisis, or waiting for an interacting group. Instructions avoid moral or partisan labels, individual ballots remain private, and only group-level outcomes are shown to other participants. The debrief explains the research interpretation only after paid decisions and emphasizes that no individual choice is treated as evidence of corruption, authoritarianism, or opposition to democracy.

Participant compensation is EUR 10–15, with a ceiling of EUR 4,500 for 300 completed participants. Laboratory charges and arrangements for timely arrivals who cannot enter a valid roster follow the agreement with the host laboratory. Secure data services and routine dissemination costs follow available project funds. The experiment uses the open-source oTree platform and requires no proprietary experimental-software licence.

8 Data Governance, Schedule, and Outputs

The experimental interface uses participant codes rather than names. The production oTree study and coded research data are hosted on University of Turku IT servers in Finland, including for sessions conducted at DMLab. Each host laboratory separately administers the identifying information needed for recruitment, attendance, and payment under its own privacy notice. The research team receives only coded experimental data and never receives any temporary identity-to-code link; the host laboratory removes that link after payment under its procedure. Access to coded research data is limited to Janne Tukiainen, Hector Bahamonde, and authorised personnel. Reporting will use aggregate or de-identified data, while coded data may be retained for verification and compatible future research under the research privacy notice.

Table 3: Planned project schedule

Period	Main activities
September 2026	Ethical review, final programming and translation, technical and behavioral pilots, data-protection review, and preregistration.
1–31 October 2026	Recruitment and laboratory data collection at UTU Choice Lab and, if used, DMLab.
November–December 2026	Data verification, de-identification, reproducible cleaning, primary and secondary analyses, and robustness checks.
January–March 2027	Manuscript preparation and preparation of de-identified replication materials.
After reporting	Retention, controlled archiving, or deletion in accordance with the approved data management plan and privacy notice.

The main outputs will be academic articles, presentations, and replication materials. Consistent with University open-science policy, code, documentation, and data that can be shared safely will be made as open as possible and as restricted as necessary. The primary conclusion will follow the estimand and decision rule stated above; related econometric and randomization-based methods may be used to assess precision and robustness without changing the randomized comparison.

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